

Guan-Horng Liu

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RESEARCH INTERESTS

deep generative models, Schrödinger bridge, dynamic optimal transport, stochastic optimal control, scalable higher-order optimization, Hamilton-Jacobi Bellman principle, forward-backward stochastic differential equations.

RESEARCH EXPERIENCE

FAIR, Meta AI

Research Scientist

Research Scientist Intern

New York, NY

09/2024 - present

Summer 2023

NVIDIA Research

Research Intern

Santa Clara, CA

Summer 2022

Uber Advanced Technology Group

Robotics Research Engineer

Pittsburgh, PA

09/2017 – 12/2018

Aptiv Mobility Group

Research Intern

Pittsburgh, PA

Summer 2016

EDUCATION

Georgia Institute of Technology

Ph.D. in Machine Learning (GPA: 4.0/4.0)

◦ Advisor: Evangelos A. Theodorou

◦ Received ICLR'21 Spotlight, ICML'21 Oral, NeurIPS'21 Spotlight, and NeurIPS'22 Oral

Atlanta, GA

07/2024

Carnegie Mellon University

M.S. in Robotics (GPA: 4.0/4.0)

◦ Advisor: George Kantor

Pittsburgh, PA

05/2017

Tokyo Institute of Technology

Research Exchange Program (GPA: 4.0/4.0)

◦ Advisor: Edwardo F. Fukushima

Tokyo, Japan

06/2014

National Taiwan University

B.S. in Mechanical Engineering (GPA: 3.99/4.0)

◦ Advisor: Pei-Chun Lin

◦ Graduated Cum Laude; Best Paper Award in 2013 IEEE/SICE ISS

Taipei, Taiwan

06/2013

Last update: 06/2025

PUBLICATIONS

(*Equal contribution *Core contributors †Equal advising ‡Alphabetical order)

Preprints

- [P1] Momentum Multi-Marginal Schrödinger Bridge Matching,
P. Theodoropoulos, A. D. Saravanas, E. Theodorou[†], **G.-H. Liu[†]**,
2025.

Conference Papers

- [C1] Adjoint Sampling: Highly-Scalable Diffusion Samplers via Adjoint Matching,
A. J. Havens*, B. K. Miller*, B. Yan*, C. Domingo-Enrich, A. Sriram, D. S. Levine, B. M. Wood,
B. Hu, B. Amos, B. Karrer, X. Fu*, **G.-H. Liu***, Ricky T. Q. Chen*,
International Conference on Machine Learning (ICML), 2025.
- [C2] Feedback Schrödinger Bridge Matching, **[Oral, 1.8%]**
P. Theodoropoulos, N. Komianos, V. Pacelli, **G.-H. Liu[†]**, E. Theodorou[†],
International Conference on Learning Representations (ICLR), 2025.
- [C3] Generalized Schrödinger Bridge Matching,
G.-H. Liu, Y. Lipman, M. Nickel, B. Karrer, E. Theodorou, Ricky T. Q. Chen,
International Conference on Learning Representations (ICLR), 2024.
- [C4] A Robust Differential Neural ODE Optimizer,
P. Theodoropoulos, **G.-H. Liu**, T. Chen, A. D. Saravanas, E. Theodorou,
International Conference on Learning Representations (ICLR), 2024.
- [C5] Mirror Diffusion Models for Constrained and Watermarked Generation,
G.-H. Liu, T. Chen, E. Theodorou[†], M. Tao[†],
Advances in Neural Information Processing Systems (NeurIPS), 2023.
- [C6] Deep Momentum Multi-Marginal Schrödinger Bridge,
T. Chen, **G.-H. Liu**, M. Tao, E. Theodorou,
Advances in Neural Information Processing Systems (NeurIPS), 2023.
- [C7] I²SB: Image-to-Image Schrödinger Bridge,
G.-H. Liu, A. Vahdat, D.-A. Huang, E. Theodorou, W. Nie[†], A. Anandkumar[†],
International Conference on Machine Learning (ICML), 2023.
- [C8] Deep Generalized Schrödinger Bridge, **[Oral, 1.9%]**
G.-H. Liu, T. Chen*, O. So*, E. Theodorou,
Advances in Neural Information Processing Systems (NeurIPS), 2022.
- [C9] Likelihood Training of Schrödinger Bridge using Forward-Backward SDEs Theory,
T. Chen*, **G.-H. Liu***, E. Theodorou,
International Conference on Learning Representations (ICLR), 2022.
- [C10] Second-Order Neural ODE Optimizer, **[Spotlight, 3.0%]**
G.-H. Liu, T. Chen, E. Theodorou,
Advances in Neural Information Processing Systems (NeurIPS), 2021.
- [C11] Dynamic Game Theoretic Neural Optimizer, **[Long talk, 3.0%]**
G.-H. Liu, T. Chen, E. Theodorou,
International Conference on Machine Learning (ICML), 2021.
- [C12] Differential Dynamic Programming Neural Optimizer, **[Spotlight, 3.8%]**
G.-H. Liu, T. Chen, E. Theodorou,
International Conference on Learning Representations (ICLR), 2021.

- [C13] Variational Inference MPC using Tsallis Divergence,
Z. Wang*, O. So*, J. Gibson, B. Vlahov, M. S. Gandhi, **G.-H. Liu**, E. Theodorou,
Robotics: Science and Systems (RSS), 2021.
- [C14] Learning End-to-end Multimodal Sensor Policies for Autonomous Navigation,
G.-H. Liu, A. Siravuru, S. Prabhakar, M. Veloso, G. Kantor,
Conference on Robot Learning (CoRL), 2017.
- [C15] Autonomous Control of the WAM-V Catamaran Type USV: Propulsion System Design,
G.-H. Liu, A. Y. Yasutomi, A. Holgado, E. F. Fukushima,
Annual Conference of the Robotics Society of Japan, 2014.
- [C16] Design of a Kangaroo Robot with Dynamic Jogging Locomotion, [Best Paper Award, 0.8%]
G.-H. Liu, H.-Y. Lin, H.-Y. Lin, S.-T. Chen, P.-C. Lin,
IEEE/SICE International Symposium on System Integration (ISS), 2013.

Journal Papers

- [J1] Optimal Transport for Generating Transition States in Chemical Reactions,
C. Duan*, **G.-H. Liu***, Y. Du*, T. Chen, Q. Zhao, H. Jia, C. P. Gomes, E. Theodorou, H. J. Kulik,
Nature Machine Intelligence, 2025.
- [J2] Improving Generative Model-based Unfolding with Schrödinger Bridges,
S. Diefenbacher[‡], **G.-H. Liu[‡]**, V. Mikuni[‡], B. Nachman[‡], W. Nie[‡],
Physical Review D, 2024.
- [J3] A Bio-Inspired Hopping Kangaroo Robot with an Active Tail,
G.-H. Liu, H.-Y. Lin, H.-Y. Lin, S.-T. Chen, P.-C. Lin,
Journal of Bionic Engineering, 2014.

Workshop Papers & Technical Reports

- [O1] Large-Scale Optimization for DNN Architecture: A Dynamical System Theory,
G.-H. Liu,
Georgia Tech Machine Learning Ph.D. Thesis, 2024.
- [O2] Augmented Bridge Matching,
V. D. Bortoli, **G.-H. Liu**, T. Chen, E. Theodorou, W. Nie,
Preprint, 2023.
- [O3] Improved Sampling via Learned Diffusions,
L. Richter*, J. Berner*, **G.-H. Liu**,
ICML Workshop on New Frontiers in Learning, Control, Dynamical Systems, 2023.
- [O4] Spatio-Temporal Differential Dynamic Programming for Control of Fields,
E. N. Evans, O. So, A. P. Kendall, **G.-H. Liu**, E. Theodorou,
Preprint, 2021.
- [O5] Deep Learning Theory Review: An Optimal Control and Dynamical Systems Perspective,
G.-H. Liu, E. Theodorou,
Preprint, 2019.
- [O6] High Dimensional Planning and Learning for Off-Road Driving,
G.-H. Liu,
CMU Robotics Institute Master Thesis, 2017.

PATENTS

Conditional Diffusion Model for Data-to-Data Translation

U.S. Patent Application No. 18/431,527 08/2024

HONORS & AWARDS

Fellowships & Scholarships

AE Graduate Research Fellowship, Georgia Tech 2022 – 2023
Study Abroad Scholarship, Ministry of Education, Taiwan 2019 – 2021
Student Exchange Scholarship, JASSO, Japan 2013

Awards & Prizes

Best Paper Award, IEEE/SICE ISS 2013
Third Prize, Chuian-Yan Thesis Paper Competition, Taiwan 2013
Presidential Awards ($\times 4$), Top 5% in National Taiwan University 2009 – 2014

INVITED TALKS

Learning Scalable Diffusion Models using Optimality and Constraint Structures

Learning on Graphs NYC Meetup 11/2024
CS5785 Applied Machine Learning, Cornell Tech (Host: Brandon Amos) 09/2024
AMLab, University of Amsterdam 05/2024
Sony AI 04/2024
National Taiwan University (Host: Shao-Hua Sun) 04/2024
Appier Group Inc. 01/2024
School of Industrial and Systems Engineering, Georgia Tech (Host: Yao Xie) 12/2023
FAIR, Meta AI 11/2023
Nvidia Research 10/2023

Mirror Diffusion Models

Learning on Graphs and Geometry Reading Group 10/2023

(Generalized) Schrödinger Bridge

AE8803 Optimal Transport Theory & Applications, Georgia Tech (Host: Yongxin Chen) ... 04/2025
Learning on Graphs and Geometry Reading Group 02/2023
NeurIPS Workshop on Score-Based Methods 12/2022
Rough Path Interest Group, Alan Turing Institute 11/2022
IBM Research Seminar 11/2022
School of Mathematics, Georgia Tech (Host: Molei Tao) 11/2021

Optimal Control Theoretic Neural Optimizer

AE4803 Robotic Systems and Autonomy, Georgia Tech (guest lecture) 10/2022
Rough Path Interest Group, Alan Turing Institute 12/2021
Machine Learning PhD Seminar, Georgia Tech (contributed talk) 10/2021
NeurIPS Workshop on Optimization for Machine Learning (spotlight talk) 12/2020

ACADEMIC SERVICES

Co-organizer: ICML 2024 Workshop on [Structured Probabilistic Inference & Generative Modeling](#)
ICML 2023 Workshop on [New Frontiers in Learning, Control, and Dynamical Systems](#)

Area Chair: NeurIPS 2023 Workshop on [AI for Scientific Discovery: From Theory to Practice](#)

Reviewer: ICLR (2024 – 2025), ICML (2023 – 2025), NeurIPS (2023 – 2025), L4DC (2023 – 2024),
IJCAI (2024), AISTATS (2025)